

Ductile damage modelling of heterogeneous materials using a two-scale computational approach

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Abstract

The paper presents a new two-scale methodology for modelling of damage in ductile heterogeneous materials employing homogenization approach over the two microstructural models. The first microstructural model comprises elastoplastic behaviour without softening, while only microstructural softening is homogenized in the second model. The scale transition of the state variables is based on the first-order computational homogenization scheme. The microstructural damage response is governed by the nonlocal variable embedded into the gradient elastoplastic formulation using the implicit localization approach. The averaging computation over the microstructural softening zones is driven by the damage variable and the local equivalent strain. Therein discretization is performed by using the mixed finite element formulation including the nonlocal equivalent plastic strain interpolation, while the standard displacement based finite elements, employing the simple constitutive relation with damage variable, are used at the macrolevel. The capability of modelling material damage responses at both the micro and macroscale is demonstrated by the benchmark examples.

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1. Introduction

In recent years, modern structures are becoming more complex, and the requirements for safety, reliability, maintainability and efficiency represent an increasing challenge in the scientific and engineering community. In addition, the load-carrying capacity of the statically and dynamically loaded structures decreases continuously during their lifetime due to various degradation mechanisms, usually appearing at microlevel, such as material aging, corrosion, fatigue and structural damage. These undesired phenomena can result in loss of structural integrity, which can have catastrophic consequences. Unfortunately, the concept of classical constitutive modelling of material behaviour is unable to describe the microstructural deformation responses. Therefore, the need for a modelling of evolving microstructure has been accounted for in the framework of multiscale methods relying on the homogenization concepts [1–5]. The mostly applied computational homogenization allows incorporation of the microstructure into a standard continuum model turning standard boundary value problem into a nested boundary value problem, containing both the macroscale and the microscale level [6,7]. In such scheme, the

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macroscopic material model is provided by a locally attached microscopic boundary value problem driven by the macroscopic quantities. The macroscopic properties are determined by the homogenization process acting on an effective, homogenized sample of material called representative volume element (RVE) [8,9].

Over the past decades, extensive studies on numerical methods for modelling of failure processes have been performed. Unfortunately, usage of the conventional computational homogenization framework for simulation of the softening responses of heterogeneous materials seems to be limited, due to questionable representativeness of the RVE [10,11] and the macroscopic boundary value problem can become ill-posed [12]. The other well-known issue is the lack of separation of scales [13,14]. To resolve these issues, the method of computational homogenization of interfacial volumes towards cohesive zones has been developed in [15], resolving the localization at the macroscopic scale. Another class of methods dealing with the localization at the macroscale are continuous-discontinuous approaches, where the microscale properties within the localization band are directly upscaled into the macroscale enrichment, representing the fracture. In this case the macrostructure can be enriched by employing the weak discontinuity [16] as well as the strong discontinuity [17,18]. Furthermore, in recent studies, the coupled volume method [12] has also shown high versatility. In the contributions [19,20], the failure zone averaging scheme has been proposed, as an extension of the developments presented in [18]. The so-called Failure-Oriented Multi-scale variational Formulation (FOMF) approach, which consistently resolves the failure of heterogeneous materials by incorporating Non-Standard Boundary Conditions is discussed in [21,22]. In all publications mentioned above, only quasi-brittle material behaviour is taken into consideration.

An approach which copes with the problem of ductile softening in heterogeneous materials is presented in [23], where the solution method for strong coupling of two scales is proposed, employing the implicit integration schemes at each scale. Herein exploiting of the principle of the operator split method ensures admissibility of the state variables. The two-phase material is considered, where one phase exhibits nonlinear behaviour due to plasticity, while the other phase is represented by a damage model. Recently, phase-field modelling of fracture response in elastic solids became a well-established framework that overcomes the limitations of the classical Griffith theory in the prediction of crack nucleation and identification of complicated crack paths. In a paper [24], an extension of phase-field formulation towards ductile fracture in elastoplastic solids is proposed, assuming the quasi-static kinematically linear regime. In the model thermodynamical consistency is ensured by appropriate coupling the evolution of the crack phase field with the accumulation of the plastic strains. An efficient fully three-dimensional homogenization-based numerical approach employing a dynamic link between the two scales for modelling degradation and failure in polycrystalline materials is proposed in [25]. Therein, to avoid pathological damage localization at the macro-scale, integral non-local counterparts of the strains are used. Furthermore, the homogenization scheme for calculating effective macroscopic properties of ductile damaged materials comprising mean field homogenization method is derived in [26], assuming large strain formulation. Herein, the effective energy density function is expressed by the energy density function of matrix material and the damage variable. The relation between macroscale stress and the microscale stress of the matrix is derived based on hyper-elastic model. The homogenization procedure is based on the multiplicative decomposition of the deformation gradient on volumetric and isochoric parts. However, additional assumptions are required to obtain the relation between the macroscopic and microscopic deformation gradient. In the contribution [27] an extended approach to the mean field homogenization is proposed, where the modelling of ductile damage behaviour in composite materials by means of the incremental micromechanics scheme is developed. Here to prevent too stiff macroscopic stress-strain response, an isotropization procedure is applied. On the other hand, the intensive damage localization problem is not addressed. A distinct approach has been used in [28], where the macroscopic analysis of ductile failure in porous ductile materials containing brittle inclusions is modelled by the homogenization-based continuum plasticity-damage model. Herein, the modified anisotropic Gurson-Tvergaard-Needleman (GTN) elastoplastic model for porous ductile materials is assumed. The model parameters are calibrated using the homogenization performed on the RVEs by means of the locally enriched Voronoi cell finite element (FE) model. Unfortunately, the model seems to be limited for the highly localized damage evolution. The previously mentioned FOMF formulation [21] has also been used for modelling of the ductile damage in [29]. Unfortunately, the research is suited only for the early damage stages, accounting for plastic deformation and matrix-nodule debonding. Similarly, a numerical analysis of the ductile damage behaviour of the microporous alumina was carried out by means of the full-field homogenization approach based on the FE model extracted from scanning electron microscope (SEM) images in [30]. But again, the extensive strain localization is not simulated. More details and the review on recent developments in the multiscale modelling of ductile damage

responses can be found in [31–33]. Despite a lot of efforts, an efficient method which can resolve the problem of consistent upscaling material behaviour during softening, especially for the materials exhibiting plastic yielding, is not available yet.

In the present paper, a new two-scale approach for modelling of ductile damage in heterogeneous materials is derived. The procedure relies on the principles of the damage mechanics, where the effective constitutive relation comprises the stiffness of undamaged material and appropriate softening law. Here, the authors propose the same constitutive assumption at the macroscale, but instead of analytical relations, the constitutive responses of both the undamaged material, as well as the softening part are obtained by the averaging procedure performed separately on the microstructural models. The first-order computational homogenization is applied. The standard von Mises J_2 elastoplastic model is applied for modelling undamaged microstructural deformation responses. The implicit gradient enhanced elastoplasticity, also employing the von Mises yield function is applied for the consideration of the softening behaviour. The averaging procedure during the microstructural softening evolution is driven by the damage variable and the local equivalent strain, which ensures a more realistic localization area. The benchmark examples show the realistic softening evolution at microlevel as well as the macroscopic localization band formation up to the complete failure.

The paper is organized as follows. In Section 2 the ductile damage model governed by the nonlocal equivalent plastic strain measure is discussed, and its implementation into the finite element framework is described. Section 3 deals with the finite element formulation adopted for the discretization of the macrostructure, which includes the damage variable in the constitutive model. In this section the basic finite element relations are presented. In Section 4, the new scale transition scheme is derived and verified. The conventional first-order homogenization is just briefly presented, while particular attention is directed to the newly proposed procedure for determination of the softening band at the microscale, where averaging of the damage variable is conducted. The new multiscale scheme for modelling of ductile damage in heterogeneous materials is tested in Section 5 by means of the usual benchmark problems of the weakened strip subjected to the tension and plate subjected to compressive loading. In Section 6 concluding remarks are drawn.

2. Nonlocal implicit ductile damage model

It is generally known that conventional numerical approaches are unable to adequately solve boundary value problem when material exhibits softening behaviour. More precisely, partial differential equations associated with a material damage become ill-posed, resulting in their loss of ellipticity. As a consequence, damage zone tends to localize into a band of zero width, and the results obtained show dependence on the mesh refinement and alignment [34]. From physical and numerical standpoint, the nonlocal continuum models have shown very good performance in solving problems related to the material softening [35,36]. The implicit approach has also become very popular in the literature for modelling brittle [37,38], as well as for ductile damage [39–41]. In this method, an additional differential equation, which relates local and nonlocal variables should be solved. For problems with softening behaviour in ductile materials, the nonlocal variable is represented by the nonlocal equivalent plastic strain.

In this work, the nonlocal implicit ductile damage model, derived in [39] has been employed for modelling of the ductile damage at the microstructural level. Herein, the damage model along with its finite element implementation will be briefly summarized, for the sake of clarity. The damage variable is governed by the nonlocal equivalent plastic strain measure $\bar{\varepsilon}_p$, introduced by an additional differential equation, as described above. In the yield function, the standard hardening rule is employed, where due to softening influence the yield stress $\sigma_y(\varepsilon_p)$ is reduced in a multiplicative way by a scalar damage variable $D(\bar{\varepsilon}_p)$ as

$$F(\boldsymbol{\sigma}, \varepsilon_p, D) = \sigma_e(\boldsymbol{\sigma}) - (1 - D) \sigma_y(\varepsilon_p). \quad (1)$$

In Eq. (1), σ_e represents the equivalent Von Mises stress, while ε_p denotes the local equivalent plastic strain measure. For the implementation of the presented damage model into the finite element method, the C^0 continuous 4-node quadrilateral finite element for plane strain conditions has been employed. To incorporate gradient-enhanced elastoplastic framework, besides the standard equilibrium equation, an additional partial differential equation of the Helmholtz type is solved

$$\bar{\varepsilon}_p - l^2 \nabla^2 \bar{\varepsilon}_p = \varepsilon_p. \quad (2)$$

In Eq. (2), the regularizing microstructural parameter l^2 appears, which is related to the characteristic size of the underlying microstructure. Beside the implicit form (2), the explicit approaches are available in the literature [42,43]. However, they require C^1 interpolation employment and experience difficulties in satisfying additional boundary condition

$$(\nabla \otimes \bar{\varepsilon}_p) \cdot \mathbf{n} = 0 \quad (3)$$

with $\mathbf{n}$ as outward normal, which is supposed to be prescribed on the moving elastoplastic boundary (EPB). In the implicit approach, this issue is conveniently overwhelmed, because the condition (3) should be prescribed at the external boundary only. To find the solution of the differential equation (2), the additional nodal degree of freedom $\bar{\varepsilon}_p$ is required, which consequently leads to the mixed finite element formulation derived from the following expressions

$$\begin{aligned} \int_V \delta \boldsymbol{\varepsilon}^T \boldsymbol{\sigma} dV &= \int_A \delta \mathbf{u}^T \mathbf{t} dA, \\ \int_V \mathbf{N} \bar{\varepsilon}_p dV + l^2 \int_V \nabla \mathbf{N} \nabla \bar{\varepsilon}_p dV &= \int_V \mathbf{N} \varepsilon_p dV. \end{aligned} \quad (4)$$

Herein the first equation represents the well-known form of the principle of the virtual work, and the second one stands for the weak form of the implicit formulation of the nonlocal equivalent plastic strain. In the second equation of (4) $\mathbf{N}$ is the matrix of the element shape functions. Based on the consistent linearization, as explained in [39], the following constitutive updates are derived

$$\begin{aligned} \Delta \boldsymbol{\sigma} &= \mathbf{C}_{\sigma \varepsilon} : \Delta \boldsymbol{\varepsilon} - \mathbf{C}_{\sigma \bar{\varepsilon}_p} \Delta \bar{\varepsilon}_p, \\ \Delta \varepsilon_p &= \mathbf{C}_{\varepsilon_p \varepsilon} : \Delta \boldsymbol{\varepsilon} - \mathbf{C}_{\varepsilon_p \bar{\varepsilon}_p} \Delta \bar{\varepsilon}_p \end{aligned} \quad (5)$$

with the constitutive operators $\mathbf{C}_{\sigma \varepsilon}$, $\mathbf{C}_{\sigma \bar{\varepsilon}_p}$, $\mathbf{C}_{\varepsilon_p \varepsilon}$, $\mathbf{C}_{\varepsilon_p \bar{\varepsilon}_p}$ correlating the stress and local equivalent plastic strain to the strain tensor and nonlocal strain. By means of the usual mathematical manipulations the finite element equation is derived in the following form

$$\begin{pmatrix} \mathbf{K}_{uu} & \mathbf{K}_{u\bar{\varepsilon}_p} \\ \mathbf{K}_{\bar{\varepsilon}_p u} & \mathbf{K}_{\bar{\varepsilon}_p \bar{\varepsilon}_p} \end{pmatrix} \begin{pmatrix} \Delta \mathbf{u} \\ \Delta \bar{\varepsilon}_p \end{pmatrix} = \begin{pmatrix} \mathbf{F}_e^u - \mathbf{F}_i^u \\ \mathbf{F}_i^{\bar{\varepsilon}_p} \end{pmatrix}. \quad (6)$$

In Eq. (6), indexes used in the stiffness matrix of the element denote the submatrices related to the displacements, u , and the nonlocal equivalent plastic strain $\bar{\varepsilon}_p$. The particular stiffness submatrices are:

$$\begin{aligned} \mathbf{K}_{uu} &= \int_V \mathbf{B}^T \mathbf{C}_{\sigma \varepsilon} \mathbf{B} dV, \\ \mathbf{K}_{u\bar{\varepsilon}_p} &= - \int_V \mathbf{B}^T \mathbf{C}_{\sigma \bar{\varepsilon}_p} \mathbf{N} dV, \\ \mathbf{K}_{\bar{\varepsilon}_p u} &= - \int_V \mathbf{N}^T \mathbf{C}_{\varepsilon_p \varepsilon} \mathbf{B} dV, \\ \mathbf{K}_{\bar{\varepsilon}_p \bar{\varepsilon}_p} &= \int_V (\mathbf{N}^T (1 - \mathbf{C}_{\varepsilon_p \bar{\varepsilon}_p}) \mathbf{N} + l^2 \mathbf{B}^T \mathbf{B}) dV, \end{aligned} \quad (7)$$

where $\mathbf{B}$ is the standard matrix of the shape function derivatives. The nodal forces of the element are

$$\begin{aligned} \mathbf{F}_i^u &= \int_V \mathbf{B}^T \boldsymbol{\sigma} dV, \\ \mathbf{F}_i^{\bar{\varepsilon}_p} &= \mathbf{K}_{\bar{\varepsilon}_p \bar{\varepsilon}_p} \bar{\varepsilon}_p - \int_V \mathbf{N}^T \varepsilon_p dV, \\ \mathbf{K}_{\bar{\varepsilon}_p} &= \int_V (\mathbf{N}^T \mathbf{N} + l^2 \mathbf{B}^T \mathbf{B}) dV. \end{aligned} \quad (8)$$

In the examples presented in the paper, the exponential damage law will be used, in the form

$$D(\bar{\varepsilon}_p) = 1 - e^{-\beta \bar{\varepsilon}_p}, \quad (9)$$

where β represents the material dependent softening exponent, defining slope of the softening curve. Considering the damage law (9), the constitutive fourth-order tensor relating the stress and the strain can be written as

$$\mathbf{C}_{\sigma \varepsilon} = \mathbf{C}^* - \frac{\mathbf{C}^* \mathbf{n} \mathbf{n}^T \mathbf{C}^*}{h(1 - D) + 3G}, \quad (10)$$

where h defines the hardening modulus, G is the shear modulus, and in this case $\mathbf{n}$ denotes the normal to the yield surface, $\mathbf{n} = \frac{\partial F}{\partial \boldsymbol{\sigma}}$. The tensor $\mathbf{C}^*$ is expressed in the form

$$\mathbf{C}^* = \left(\mathbf{C}_{\text{el}}^{-1} + \frac{\Delta \gamma}{\sigma_e} (\mathbf{m} - \mathbf{m}^T) \right)^{-1}. \quad (11)$$

In (11), $\mathbf{C}_{\text{el}}$ is the Hookean elasticity tensor, $\Delta \gamma$ is the plastic multiplier and $\mathbf{m}$ is defined as

$$\mathbf{m} = \frac{3}{2} \left({}^4\mathbf{I} - \frac{1}{3} \mathbf{II} \right). \quad (12)$$

In relation (12), ${}^4\mathbf{I}$ is the fourth-order unit tensor, while $\mathbf{I}$ is the second-order unit tensor. The constitutive second-order tensor relating the stress to the nonlocal equivalent plastic strain is expressed as

$$\mathbf{C}_{\sigma \bar{\varepsilon}_p} = \frac{\sigma_y \beta (1 - D)}{h(1 - D) + 3G} \mathbf{C}^* \mathbf{n}. \quad (13)$$

The similar form is preserved in the constitutive tensor relating the local equivalent plastic strain to the strain tensor

$$\mathbf{C}_{\varepsilon_p \varepsilon} = \frac{1}{h(1 - D) + 3G} \mathbf{n}^T \mathbf{C}^*. \quad (14)$$

Finally, correlation among the local and the nonlocal equivalent plastic strain is derived by the scalar value as

$$C_{\varepsilon_p \bar{\varepsilon}_p} = \frac{\sigma_y \beta (1 - D)}{h(1 - D) + 3G}. \quad (15)$$

The nonlocal formulation shown here has advantages from numerical and theoretical aspects in comparison to other available formulations [34,44]. However, the spurious damage growth can occur, as the consequence of the nonlocal variable appearing in the entire numerical model. This gives the possibility of the energy transfer between the elastically unloading region and the localization zone, which results in the smeared damage zone instead in the localized deformation band. For more details, see [45,46]. All algorithms derived have been implemented into the FE software ABAQUS via user subroutine UEL [47].

3. Macroscale finite element formulation

At the macroscale the standard 4-node quadrilateral finite element is used, where the small strain assumption is adopted. The element contains the bilinear displacement field approximation and the two displacements per node. The reduction of the material stiffness involved into the standard finite element formulation is expressed as

$$\mathbf{C} = (1 - D) \mathbf{C}_B. \quad (16)$$

In Eq. (16), $\mathbf{C}$ is the material matrix describing effective material behaviour and $\mathbf{C}_B$ denotes the constitutive matrix of the bulk material of a heterogeneous structure. Hence, at the macroscale, the basic stress relation of damage mechanics

$$\boldsymbol{\sigma} = (1 - D) \mathbf{C}_B \boldsymbol{\varepsilon} \quad (17)$$

is assumed, where $\boldsymbol{\sigma}$ represents the effective stress tensor. According to the incremental formulation, the updates of the strain tensor and the damage variable in the time interval (t^{i-1}, t) have the form

$$\begin{aligned} \boldsymbol{\varepsilon} &= \boldsymbol{\varepsilon}^{i-1} + \Delta \boldsymbol{\varepsilon}, \\ D &= D^{i-1} + \Delta D. \end{aligned} \quad (18)$$

The incremental update of the effective stress may be computed as

$$\Delta \boldsymbol{\sigma} = (1 - D^{i-1}) \mathbf{C}_B \Delta \boldsymbol{\varepsilon} - \Delta D \mathbf{C}_B \boldsymbol{\varepsilon}^{i-1}, \quad (19)$$

where the following incremental change of the damage variable is

$$\Delta D = \left(\frac{dD}{d\boldsymbol{\varepsilon}} \right)^{i-1} \Delta \boldsymbol{\varepsilon}. \quad (20)$$

Using the relation above the incremental principal of the virtual work may be expressed as

$$\int_V \left[(\delta \boldsymbol{\epsilon})^T (\boldsymbol{\sigma}^{i-1} + \Delta \boldsymbol{\sigma}) \right] dV = \int_A \delta (\Delta \mathbf{u})^T \mathbf{t} dA, \quad (21)$$

which after discretization procedure yields

$$\int_V \left[(\mathbf{B} \delta(\Delta \mathbf{v}))^T \boldsymbol{\sigma}^{i-1} \right] dV + \int_V \left[(\mathbf{B} \delta(\Delta \mathbf{v}))^T \Delta \boldsymbol{\sigma} \right] dV = \int_A (\mathbf{N} \delta(\Delta \mathbf{v}))^T \mathbf{t} dA. \quad (22)$$

Herein, $\Delta \mathbf{v}$ represents an update of the element nodal degrees of freedom. Inserting the stress update (19) into relation (22), and after usual mathematical manipulations, Eq. (22) can be rewritten as

$$\int_V (\mathbf{B}^T \mathbf{C}_M \mathbf{B}) \Delta \mathbf{v} dV = \int_A \mathbf{N}^T \mathbf{t} dA - \int_V (\mathbf{B}^T \boldsymbol{\sigma}^{i-1}) dV, \quad (23)$$

corresponding to the finite element equation

$$\mathbf{K} \Delta \mathbf{v} = \mathbf{F}_e - \mathbf{F}_i. \quad (24)$$

In Eq. (24), the stiffness matrix $\mathbf{K}$ has the standard form

$$\mathbf{K} = \int_V \mathbf{B}^T \mathbf{C}_M \mathbf{B} dV \quad (25)$$

with the constitutive matrix $\mathbf{C}_M$ expressed as

$$\mathbf{C}_M = (1 - D^{i-1}) \mathbf{C}_B - \left(\frac{dD}{d\boldsymbol{\epsilon}} \right)^{i-1} \mathbf{C}_B \boldsymbol{\epsilon}^{i-1}. \quad (26)$$

The vectors $\mathbf{F}_e$ and $\mathbf{F}_i$ on the right hand-side of Eq. (24) are the external and internal nodal force vectors, respectively, defined as

$$\begin{aligned} \mathbf{F}_e &= \int_A (\mathbf{N}^T \mathbf{t}) dA, \\ \mathbf{F}_i &= \int_V (\mathbf{B}^T \boldsymbol{\sigma}^{i-1}) dV. \end{aligned} \quad (27)$$

4. Scale transition scheme

In this chapter, a new procedure for the scale transition of the material behaviour incorporating elastoplastic response, as well as damage is discussed. The transition procedure relies on the first-order computational homogenization under small strain assumption. The homogenization is performed on two microstructural samples, one without damage (the bulk microstructural volume element — BMVE) and another with embedded the nonlocal ductile damage model (the damaged microstructural volume element — DMVE). The BMVE model gives insight into the averaged stress and constitutive tensor including elastoplastic phenomena, while the DMVE serves for extraction of the averaged damage variable. The homogenized results are combined at the macroscale into the single homogenized response of the heterogeneous microstructure exhibiting softening. The procedure works in the following way. Every integration point of the macrolevel mesh has two identical microstructural models attached (bulk and damaged). The macrolevel strain tensor $\boldsymbol{\epsilon}_M$, computed at a material point, is imposed at both microscale models in terms of the boundary displacements. After solution of both boundary value problems, the two-step homogenization is performed. From the BMVE, the bulk stress tensor and constitutive operator are extracted by the conventional approach, where the homogenization is performed over the complete model. From the DMVE the damage variable is obtained through the averaging over the zone detected by the procedure proposed, which will be explained in the subsequent section in more detail. As evident from Eq. (20), the derivative of the homogenized damage with respect to the macrolevel strain should be computed, which represents an obstacle, because their dependency cannot be explicitly expressed. Therefore, the correlation between the macrolevel damage and the strain tensor is approximated numerically. Due to nonlinearities involved during elastoplastic behaviour and softening, the multiscale algorithm should be performed in an incrementally-iterative scheme, via constitutive updates of the governing variables. After the homogenization, all state variables representing the stress, the damage and the constitutive operator are updated at the macrolevel material point according to relations (19), (18) and (26). All algorithms derived have been embedded into the FE software ABAQUS via user subroutines. The newly proposed micro–macro scale transition scheme is given in Fig. 1.

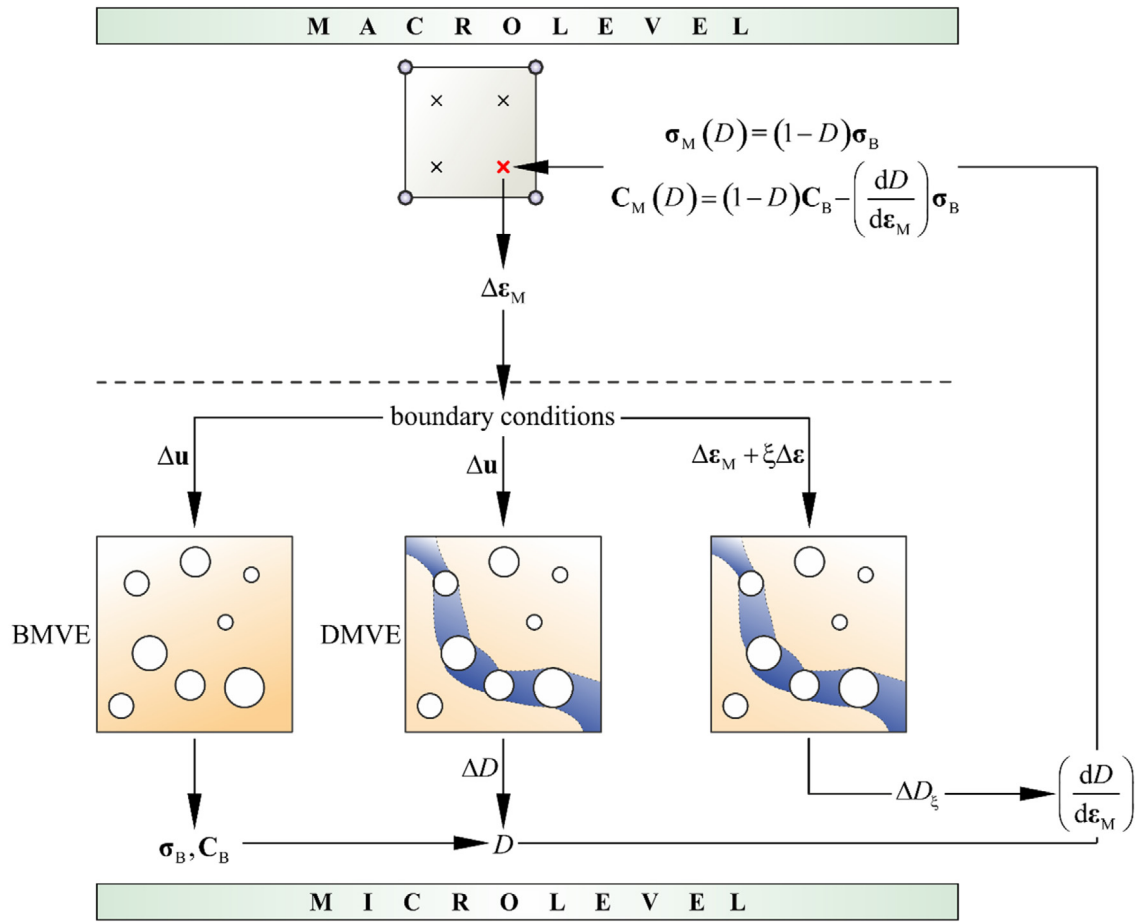


Fig. 1. Scheme of multiscale algorithm for damage modelling.

4.1. Macroscale constitutive model

As usual in multiscale numerical schemes, the constitutive behaviour at the macrolevel is unknown, and should be computed from the microstructural response. As presented above, the macroscale constitutive stiffness is described by relation (26). Obviously, the constitutive matrix of the undamaged material C_B is obtained by the conventional homogenization of the BMVE. However, the problem persists in derivation of the dependency among the damage variable and the macrolevel strain tensor $\left(\frac{dD}{d\epsilon_M}\right)^{i-1}$. There is no a convenient method to find an analytical relation for the aforementioned derivative. Therefore, the term $\frac{dD}{d\epsilon_M}$ is approximated numerically. The similar approximation for derivatives computations can be found in the available publications [1,21,48–50]. In the present contribution we start with the perturbation of the macrostrain tensor expressed as

$$\left(\xi \epsilon_{kl}^{i-1}\right)_M \mathbf{e}_k \otimes \mathbf{e}_l = \left(\epsilon_{kl}^{i-1}\right)_M \mathbf{e}_k \otimes \mathbf{e}_l + \xi \Delta \epsilon_{kl} \mathbf{e}_k \otimes \mathbf{e}_l, \quad (28)$$

where ξ is a scalar perturbation parameter, while $\Delta \epsilon_{kl}$ denotes generic strain tensor determining perturbation direction. Here the perturbation parameter is set to $\xi = 10^{-5}$. The condition (28) is imposed at the DMVE, leading to an additional boundary value problem for multiple loading cases. Considering 2D case and plane strain assumption, three combinations of indexes k and l are accounted for: 11, 22 and 12. Then the partial derivative is approximated over the localization area V_d in the form

$$\left(\frac{\partial D}{\partial (\epsilon_{kl})_M}\right)^{i-1} \approx \frac{1}{V_d} \int_{V_d} \frac{\Delta D_\xi}{\xi \Delta \epsilon_{kl}} dV_d \quad (29)$$

with ΔD_ξ representing the increment of the damage variable for the perturbation of the macrolevel strain in the direction $\xi \Delta \varepsilon_{kl}$. The localization area V_d is determined during the structural analysis of the DMVE, and it is stored in the Abaqus output database file (.odb) denoting whether particular material point is, or it is not satisfying the imposed condition, which will be considered and explained in the further section.

4.2. Microscale constitutive model

4.2.1. Homogenization of bulk microstructural volume element (BMVE)

As previously mentioned, the homogenization of the BMVE is a conventional approach, derived and used in many references [7,51,52]. Hence, only basic relations will be presented here. In the first-order homogenization, the BMVE displacement field $\mathbf{u}_m$ is computed from the macrolevel strain tensor $\boldsymbol{\varepsilon}_M$ using the relation

$$\mathbf{u}_m = \boldsymbol{\varepsilon}_M \cdot \mathbf{x} + \mathbf{r}. \quad (30)$$

In Eq. (30), $\mathbf{x}$ is the microlevel spatial coordinate and $\mathbf{r}$ represents the microfluctuation field. Since the macrolevel strain must be expressed as a volume average of its microlevel conjugate, the integral condition is imposed on the BMVE boundaries

$$\int_V (\nabla \otimes \mathbf{r}) dV = \int_{\partial V} (\mathbf{n} \otimes \mathbf{r}) d\Gamma = \mathbf{0} \quad (31)$$

with Γ representing an outer BMVE boundary. It is known that in conventional homogenization the periodic boundary conditions provide the best convergence and results. In this case the displacements of the corner nodes $\mathbf{u}_c$ are prescribed, according to the relation

$$\mathbf{u}_c = \boldsymbol{\varepsilon}_M \cdot \mathbf{x}. \quad (32)$$

The remaining BMVE boundary nodes are related accordingly to the microfluctuation periodicity assumption of the opposite edges $\mathbf{r}_L = \mathbf{r}_R$ (left–right) and $\mathbf{r}_T = \mathbf{r}_B$ (top–bottom). This leads to the periodicity equations which are imposed on the corresponding BMVE boundaries

$$\begin{aligned} \mathbf{u}_R - \mathbf{u}_L &= \boldsymbol{\varepsilon}_M \cdot (\mathbf{x}_R - \mathbf{x}_L), \\ \mathbf{u}_T - \mathbf{u}_B &= \boldsymbol{\varepsilon}_M \cdot (\mathbf{x}_T - \mathbf{x}_B). \end{aligned} \quad (33)$$

Based on the Hill–Mandel energy equivalence

$$\boldsymbol{\sigma}_B : \boldsymbol{\varepsilon}_M = \frac{1}{V} \int_V \boldsymbol{\sigma}_m : \boldsymbol{\varepsilon}_m dV, \quad (34)$$

and Eqs. (30) and (31), it is easy to prove that macrolevel bulk stress is expressed as

$$\boldsymbol{\sigma}_B = \frac{1}{V} \int_V \boldsymbol{\sigma}_m dV. \quad (35)$$

The constitutive operator of the bulk microstructure is obtained through the usual static condensation procedure in the form

$$\mathbf{C}_B = \frac{1}{V} \mathbf{D}^T \tilde{\mathbf{K}}_{bb} \mathbf{D}, \quad (36)$$

where $\tilde{\mathbf{K}}_{bb}$ denotes the condensed stiffness matrix of the BMVE, and $\mathbf{D}$ is the coordinate matrix of the BMVE boundary nodes [7].

4.2.2. Homogenization of damaged microstructural volume element (DMVE)

The next step is a novel procedure proposed for averaging of the damage spreading over heterogeneous microstructure. As already mentioned, averaging of the damage is performed on a separate microstructural model, named here DMVE. On the DMVE, the nonlocal implicit formulation of the ductile damage is adopted, which has been explained in Section 2. It is already known that the averaging over the whole microstructural model during softening gives overestimating results. It occurs due to the fact that the softening appears in a narrow zone of microstructure, while the remaining material is unloaded. Obviously, the unloading area in the microstructural model is much larger than the loading area, and the average value of these two regions will provide result closer to the

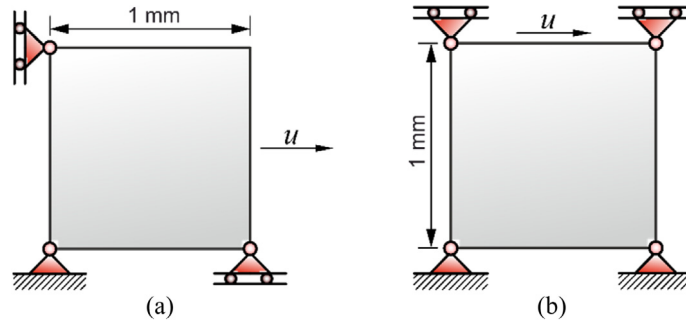


Fig. 2. Macrolevel model: (a) Mode I loading, (b) Mode II loading.

behaviour of the undamaged zone. This problem has been circumvented in [20], where only failure zone averaging has been performed in the case of brittle damage. The homogenization is performed only over the RVE zones, where material exhibits the increase in damage ($\Delta D_m > 0$), and the representativeness of the RVE is preserved. This can be easily recast into

$$D = \frac{1}{V_d} \int_{V_d} D_m dV_d, \quad (37)$$

where V_d represents localization area determined by the criterion ($\Delta D_m > 0$), D_m is damage variable computed at the microstructural scale, and D is average value upscaled to the macroscale.

However, to the authors' knowledge, there are not any publications dealing with similar approach at the microscale involving ductile damage evolution. The detection of the localization zone based only on the increase in damage, as proposed in [20], is not a sufficient condition for the averaging procedure, due to numerical issues inherent from the implicit damage formulation adopted at the DMVE. As mentioned in Section 2, energy transfer from the localization zone into unloading region can occur, resulting in the spurious damage growth and the smeared damage zone instead of localized band. The aforementioned difficulty also affects the detection of the localization zone ($\Delta D > 0$), where the macroscale average of the damage should be computed. Besides, in numerical simulations, concentrations of the nonlocal strain and accordingly damage can appear in the DMVE corners. This is partly motivated by the spurious damage growth, but also by the microfluctuation suppression at the MVE corners [53]. To exclude the unwanted regions from the averaging, an additional condition which ensures more precise capturing of the damage localization is proposed. Therefore, for a macroscale increment i , the averaging is performed at the microscale material points, where besides the criterion $\Delta D > 0$, the local equivalent strain ε_{eq} at a microlevel is greater than the threshold value ε_{eq0}^i , defined as

$$\varepsilon_{eq0}^i = \alpha \varepsilon_{eq_{max}}. \quad (38)$$

Here $\varepsilon_{eq_{max}}$ represents the maximum value of the local equivalent strain reached on whole DMVE during the loading history. The parameter α denotes a threshold parameter. The sum of all microlevel material points which fulfil the criterion (38) determines the localization area V_d . The choice of the local equivalent strain is motivated by the publications related to the quasi-brittle damage [38,46]. On the other hand, it has shown numerical stability and objectivity of the homogenized results. To demonstrate the objectivity of the localization criterion proposed, a heterogeneous macrostructure subjected to loading of mode I (tension) and mode II (shear) is considered as shown in Fig. 2. The discretization has been performed using a single finite element, to keep the scale transition problem as simple as possible. At the macrostructural material point the damage responses of the microstructural volume element presented in Fig. 3 is analysed.

The MVE size is $L = 0.2$ mm and the matrix material has Young's modulus 210 GPa and Poisson's ratio 0.3. Yield stress is 250 MPa, with hardening modulus of $h = 20,000$ MPa. For the softening behaviour, the exponential law, with microstructural parameter l equal to 0.01 mm and softening exponent $\beta = 500$, is considered. The microstructure consists of 13% porosities, with average void radius of 0.043 mm. For capturing of localization, the threshold parameter is set to $\alpha = 0.2$. The periodic boundary conditions are utilized at both, the BMVE and

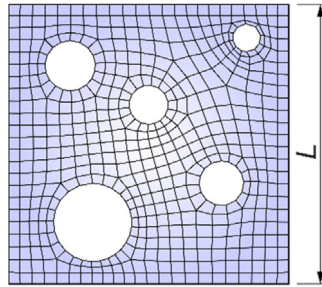


Fig. 3. MVE representing academic porous ductile material.

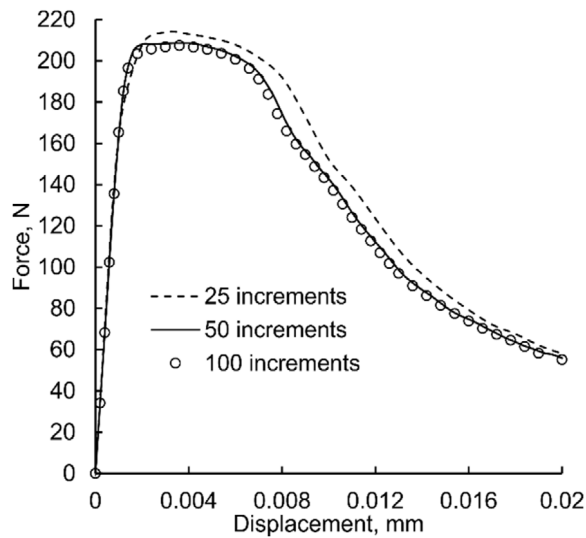


Fig. 4. Homogenized force–displacement diagram for mode I loading.

the DMVE. The three macrolevel step sizes of 25, 50 and 100 increments have been used. The macrolevel force–displacement diagrams are shown in Figs. 4 and 5. As can be seen, the homogenized solutions are independent of the macrolevel incrementation.

4.2.3. Influence of the threshold parameter α

As already mentioned in the previous section, the threshold value of the equivalent strain (38) for capturing of localization zone is expressed in terms of the parameter α . The parameter α resolves the numerical issues inherent from the nonlocal variable required in the damage model. It means, the small variations of the damage in the unloading part of the microstructure (which should not affect the average behaviour) are circumvented. On the other hand, it ensures that averaging of the damage is always performed over the loading zone with significant damage growth and accordingly with the highest increment of the strain. But, in order to achieve meaningful results, the threshold parameter should be chosen with a caution. Hence, a comprehensive study is performed in this paper. For discussion purposes, the example from the previous section is again used. The material characteristics have been preserved, only the damping parameter α has been varied. The homogenized force and damage evolution curves at the macroscale model of the loading mode I for different α are shown in Figs. 6 and 7.

As expected, the results obtained show high dependency on the value of α . With lower threshold value, the ductility is more pronounced. In Fig. 6 it is clearly displayed that the condition $\Delta D > 0$ is insufficient for determination of the localization band (curve for $\alpha = 0$). As can be seen, the standard one-step computational homogenization with averaging over entire MVE, including both undamaged and damaged areas, gives more realistic behaviour than the new procedure without threshold parameter. For the values of α below 0.2 unphysical

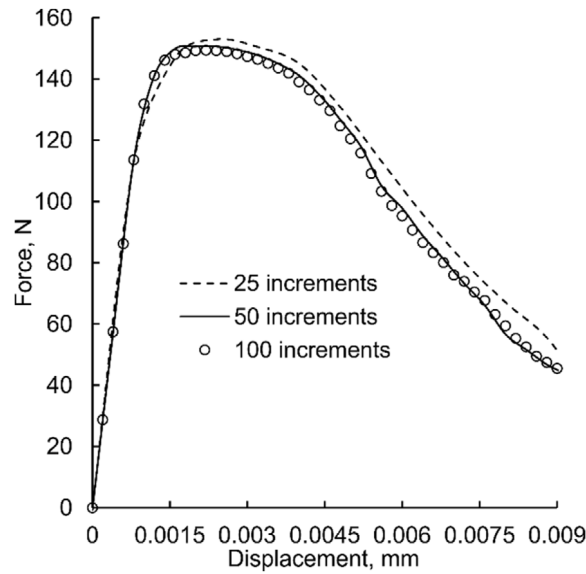


Fig. 5. Homogenized force–displacement diagram for mode II loading.

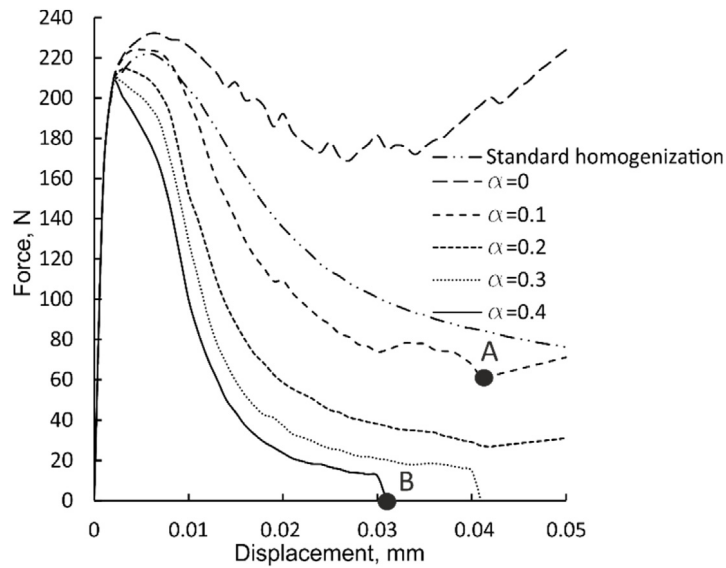


Fig. 6. Macrolevel force–displacement diagram for various values of α and Mode I loading.

change of the force, as damage is experienced. This behaviour is manifested as increase in the diagram of the force–displacement dependency (Fig. 6) and as constant value or even decrease in the diagram of the macrolevel damage–displacement dependency (Fig. 7). Clearly, such behaviour cannot be taken as realistic. As presented in Figs. 6 and 7, for $\alpha = 0.3$ and $\alpha = 0.4$ the complete failure of the microstructure is experienced. Also, it may be observed that with increase in α the dependencies of the force and damage upon displacement are showing convergent behaviour. Higher values of the threshold parameter lead to the collapse of the microstructure at lower displacement values. However, the homogenized response obtained for all variations of the threshold parameter is very similar. From Fig. 7 it can be concluded that the threshold parameter above 0.2 captures localization upon

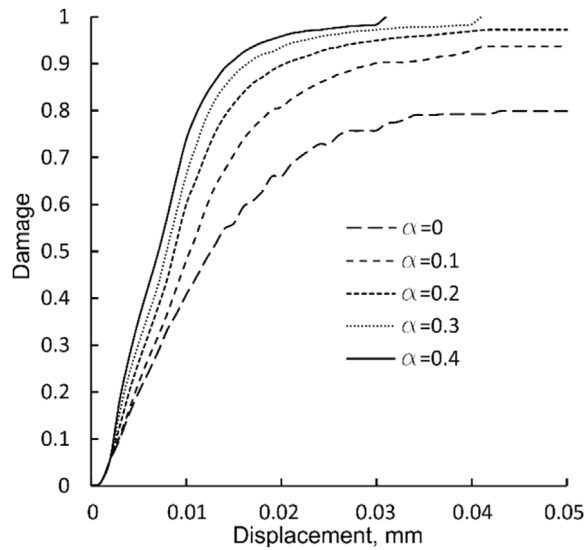


Fig. 7. Macrolevel damage–displacement diagram for various values of α and Mode I loading.

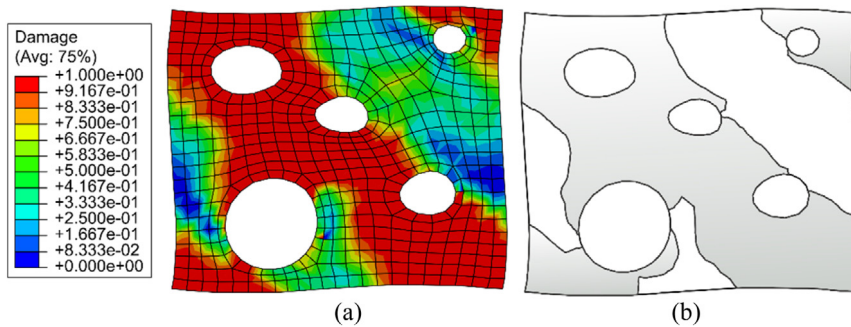


Fig. 8. DMVE at point A in Fig. 6 for $\alpha = 0.1$: (a) distribution of damage, (b) localization zone.

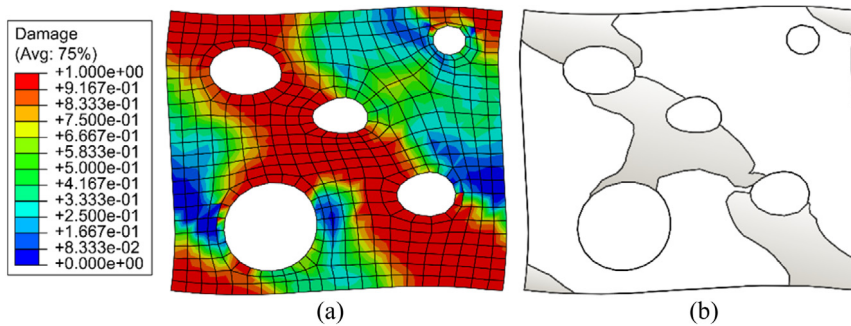


Fig. 9. DMVE at point B in Fig. 6 for $\alpha = 0.4$: (a) distribution of damage, (b) localization zone.

complete failure assigned by the unit value of the damage variable. Furthermore, the influence of the threshold parameter on the microstructural softening evolution is also shown on the DMVEs in Figs. 8 and 9.

Figs. 8 and 9 show distribution of the damage over the DMVE and the localization zone at the failure stage determined by the criterion (38), for α equal to 0.1 and 0.4. It can be noted that the damage distribution does not

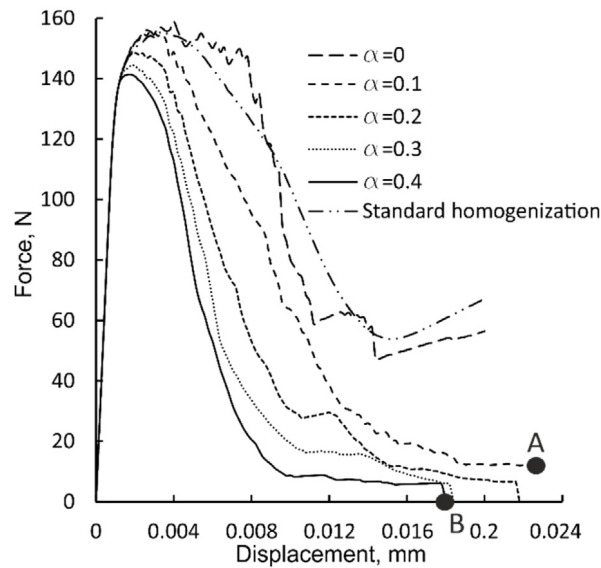


Fig. 10. Macrolevel force–displacement diagram for various values of α and Mode II loading.

depend on the value of α . Instead, the comparison of the shaded areas (part (b) of Figs. 8 and 9) demonstrates the effect of increase in the threshold parameter. Accordingly, the higher threshold parameter provides faster localization at the macroscale, due to capturing of the microstructural material points with higher values of the damage.

To show that the choice of the threshold parameter is loading-independent, shear loading computational model has also been considered. The force and damage versus displacement diagrams at the macroscale are presented in Figs. 10 and 11. As may be seen, the influence of the parameter α on the softening response is quite similar as for the case of Mode I loading. It is again demonstrated that the threshold parameter is essential for the exhibiting of physically realistic behaviour in the averaging procedure. The homogenized behaviour obtained for the values of the threshold parameter lower than 0.2 are again unable to consistently capture the localization at the DMVE. With increase in α , the homogenized behaviour gives smooth change of the macroscale variables considered. As in the case of Mode I loading, the convergence of the homogenized properties is achieved with increase in α . The distribution of the damage over the DMVE for $\alpha = 0.1$ and $\alpha = 0.4$ is displayed in Figs. 12 and 13. The softening pattern exhibited is again insensitive to the choice of the parameter α . It is clearly visible that the localization band is precisely captured, and with increase in α , the averaging band is narrowing. In Figs. 12 and 13(a) evolution of the shear band along the right DMVE boundary is visible, which is not initiated by the porosities in the microstructure and could be considered as spurious response. However, as evident in Figs. 12 and 13(b) the previous mentioned spurious localization is removed due to increase in α .

Based on the review of the diagrams presented in Figs. 6, 7, 10 and 11, along with the averaging zones presented in Figs. 8, 9, 12 and 13, the conclusion can be drawn that choice of the appropriate value of the parameter α should be motivated by the physical interpretation of the homogenized behaviour and determination of the localization band. As demonstrated, once determined, threshold parameter is load independent. Also, it can be generalized that threshold parameter $\alpha > 0.2$ can be taken not only as mode independent, but also material-independent property in the damage averaging procedure. This can be easily explained by the consideration of the damage distribution within softening band for any material. As known, highest values of damage, and accordingly strain are exhibited inside of the band. These values rapidly drop moving from the localization zone. This characteristic distribution of the damage and strain in softening problems is apparent for various kinds of materials. Threshold parameter used in this research simply discards material point too far from the localization zone, ensuring realistic material response at macroscale.

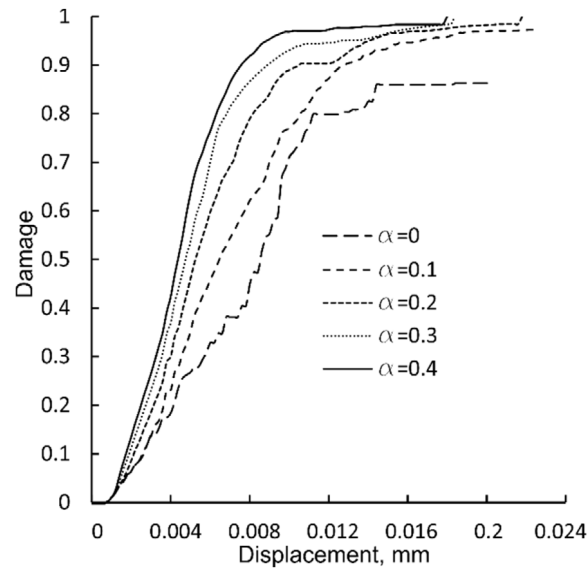


Fig. 11. Macrolevel damage–displacement diagram for various values of α and Mode II loading.

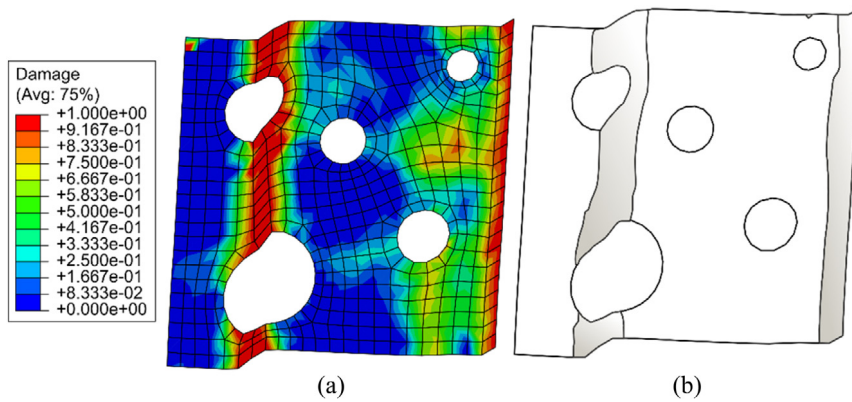


Fig. 12. DMVE at point A in Fig. 10 for $\alpha = 0.1$: (a) distribution of damage, (b) localization zone.

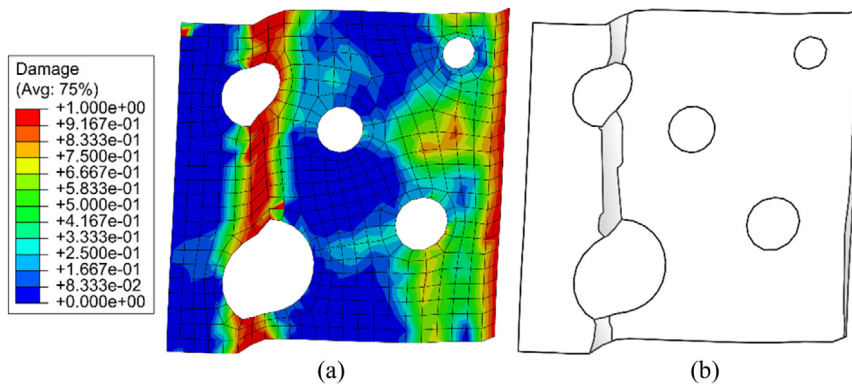


Fig. 13. DMVE at point B in Fig. 10 for $\alpha = 0.4$: (a) distribution of damage, (b) localization zone.

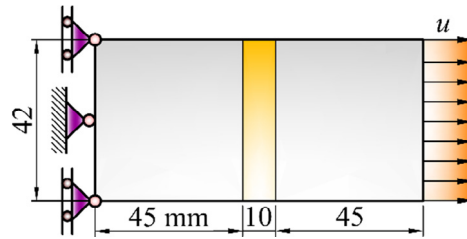


Fig. 14. Weakened strip subjected to tension.

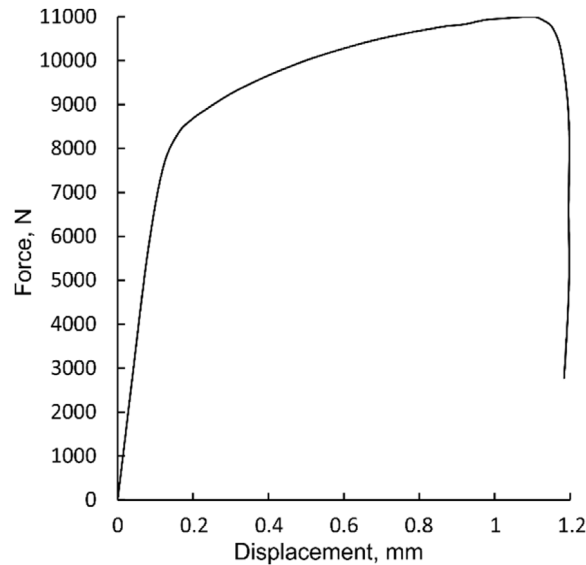


Fig. 15. Force–displacement diagram of strip subjected to tension.

5. Numerical examples

5.1. Strip subjected to tension loading

The first benchmark problem for the testing of the new two-scale procedure is a strip with a weakened zone in the middle area subjected to the tension loading. The geometry of the macromodel is shown in Fig. 14. For the discretization, 24 CPE8R finite elements [47] have been employed.

The material considered is an academic example of porous ductile material already used in the present paper, whose characteristics are described in Section 4.2, and the geometry of the MVE is presented in Fig. 3. The threshold parameter used in the example is $\alpha = 0.3$. Figs. 15 and 16 present the force–displacement diagram for the strip and the distribution of the damage on the deformed macromodel at the failure stage.

From Figs. 15 and 16 it is obvious that the new multiscale computational scheme predicts the meaningful results. The localization band has been initiated in the weakened zone and preserves typical orientation in the softening of ductile materials (45° to the loading direction). As evident, the shear band shows the tendency of localization into a narrow zone, as expected. The distribution of the damage over the DMVEs positioned at the material points A and B, marked on the deformed macromodel in Fig. 16, at the end of the loading process is presented in Fig. 17.

As Fig. 17 displays, at points A and B placed directly in the macrolevel shear band, the complete failure of the DMVE is achieved, leading to the macrolevel failure. It can be observed that the choice of the threshold parameter α allowed the development of quite broad shear band on the microstructural scale prior to complete the collapse. The DMVE associated to point C, which is placed close to the macrolevel localized area shows reasonable behaviour compared to the DMVEs at points A and B. The localization is clearly observable in Fig. 17(c), but in smaller

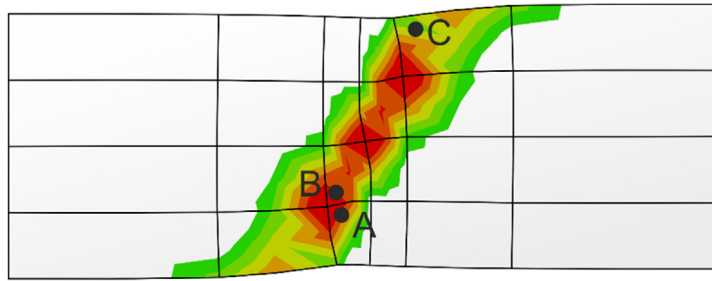


Fig. 16. Distribution of damage over macromodel.

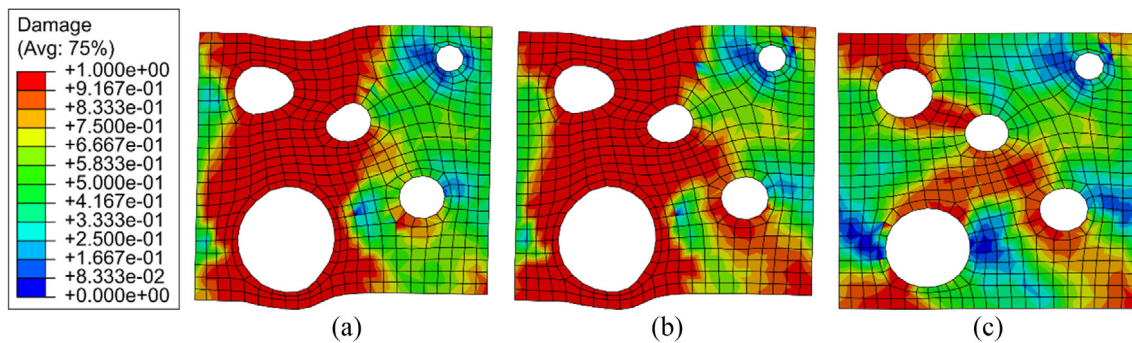


Fig. 17. Distribution of damage on DMVEs positioned at points: (a) A, (b) B and (c) C in Fig. 16.

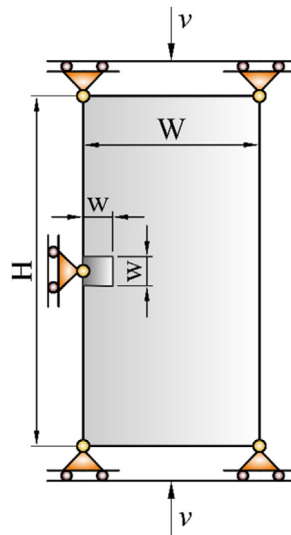


Fig. 18. Plate subjected to compressive load.

extent compared to Figs. 17(a) and (b). By overview of the results obtained, it can be concluded that the multiscale procedure proposed provides the realistic behaviour of the model, leading to the complete failure.

5.2. Shear band problem

As the second example, a plate with an imperfection subjected to compressive load is considered, as shown in Fig. 18. The dimensions of the plate are $H = 100$ mm and $W = 50$ mm. The geometry of the microstructure is the

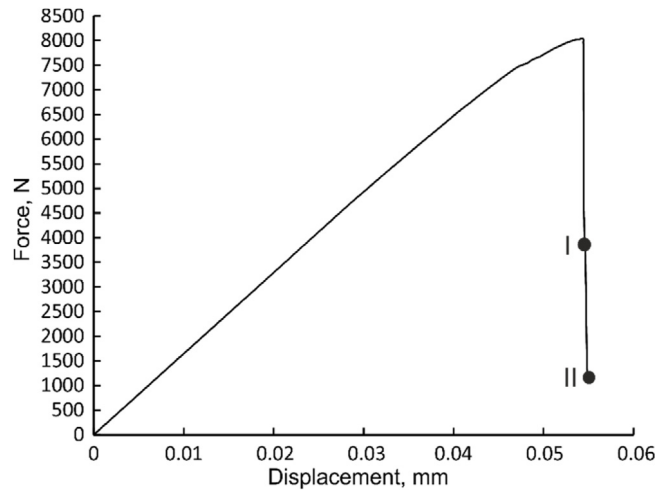


Fig. 19. Force–displacement curve of plate subjected to compressive load.

same as in the previous examples, as already presented in Fig. 3. However, the material characteristics are slightly changed. Young's modulus and Poisson's ratio of the microstructural matrix are 210 GPa and 0.3, respectively. Yield stress is 250 MPa, with hardening modulus $h = 20,000$ MPa. During softening, the exponential law is adopted, where microstructural parameter l is set to 0.1 mm and softening exponent β is 100 000. To ensure initiation of the localization, a weakened zone of $w = 5$ mm is modelled, where the yield stress of the bulk material at microstructural level is reduced to 200 MPa. Due to material heterogeneities, the structural symmetry cannot be used, and therefore the whole macrostructure is discretized by 72 CPE8R finite elements. The compressive loading is applied by means of the displacement control, where vertical displacement v is imposed on the top and bottom side of the model. Threshold parameter used for the capturing of localization at the DMVE is set to $\alpha = 0.3$.

The force–displacement curve representing the softening response is depicted in Fig. 19, while Fig. 20 presents the spreading of the softening band over the macrostructure at the stages I and II marked in Fig. 19.

As may be observed in Fig. 20(b), the distribution of the softening zones through the whole plate demonstrates the complete failure. After the damage initiation the two shear bands originating from the weakened zone are produced. However, due to the microstructural heterogeneities, which yields anisotropic response at the macrolevel, during the later loading stage only lower band spreads until the complete failure occurs. The force–displacement curve presented in Fig. 19 demonstrates relatively low ductility of the plate. It is caused by the exponential damage evolution law including high softening exponent which yields faster damage initiation. The distribution of the damage and plastic zones over the DMVEs associated to the points A and B shown in Fig. 20, for two loading stages considered, is presented in Figs. 21 and 22.

The deformation pattern over the DMVEs is in accordance to the macrolevel softening response, as expected. Both DMVEs experience almost the same failure pattern, with shear bands spread through their entire width. At the loading stage I, the DMVE associated to the point A has already exhibited the complete failure, hence the distribution of the microstructural variables is not visualized once again for the loading stage II. The DMVE of the point B shows the microstructural response only at the loading stage II, which represents the macroscale failure. As evident there is no elastoplastic response at point B of the loading stage I. Despite the relative low material ductility considered in this example, it can be seen from Figs. 21(b) and 22(b) there is significant amount of plastic strain developed prior to the macrostructural collapse.

It is useful to discuss the computational efficiency of the proposed multiscale procedure. As known, the computational homogenization scheme is time demanding by its definition, as a consequence of the microscale computations conducted “on the fly” in every macrolevel material point. In this paper, the computational efficiency is additionally tackled by multiple microscale computations required according to the algorithm presented in Fig. 1. As displayed in Fig. 1, five MVE computations should be carried out at every macrolevel material point. However, these simulations can be conducted in two stages using parallel computing. First stage consists of parallel performing

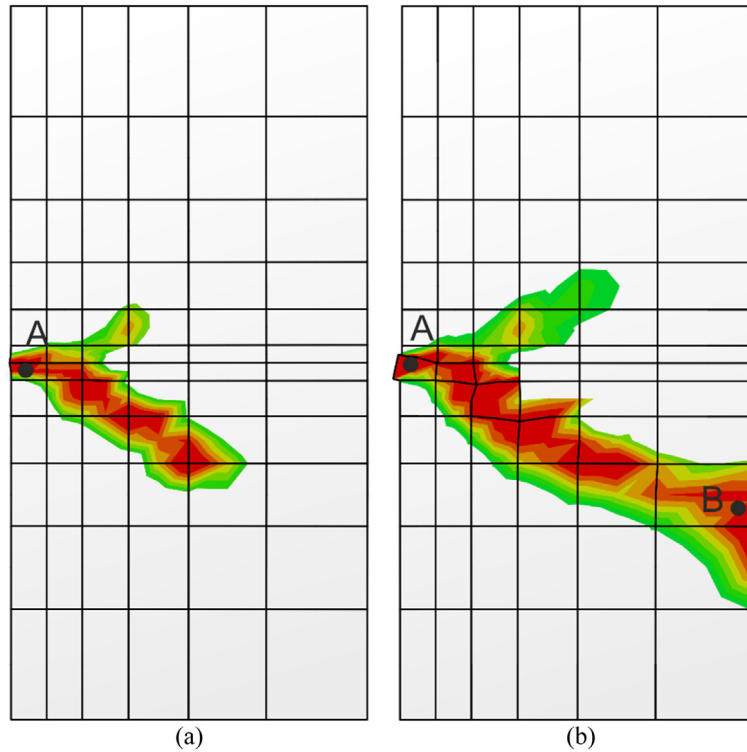


Fig. 20. Distribution of damage on plate: (a) stage I, (b) stage II.

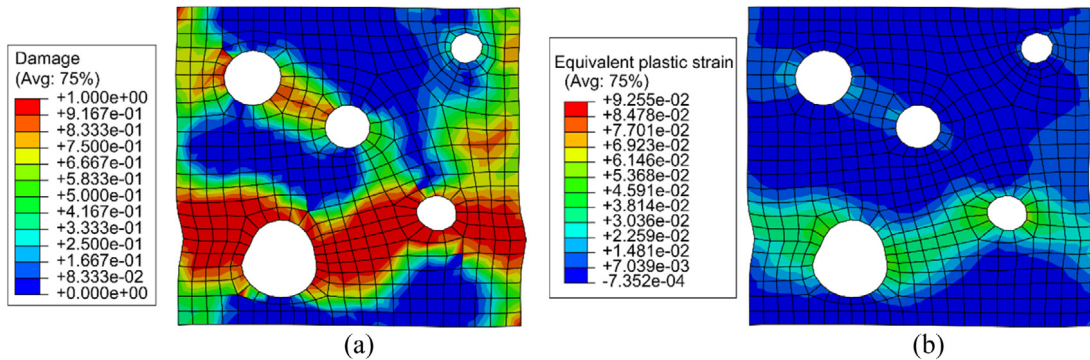


Fig. 21. Distribution of variables over DMVE at point A in Fig. 20 for loading stage I: (a) damage, (b) equivalent plastic strain.

of the BMVE and DMVE analysis. The additional speed up of those analyses can be achieved by parallelization of each particular simulation. After accomplishment of the DMVE analysis, in the second stage, three perturbation steps are imposed on the final deformed configuration of the DMVE, according to expression (28). Therein, all perturbation analyses are again computed at the same time. Due to very small loading applied, the simulations are conducted without loading incrementation, where the parallelization of each perturbation step is again possible and desirable. Besides the parallelization and boosting of the microscale computations, the parallelization of the computations at the macrolevel is also possible, where the previously described procedure is conducted in parallel for several macrolevel material points. The simulations within this paper were performed on a workstation with 48 cores and clock rate 2.2 GHz. The BMVE and DMVE analyses were carried out in parallel on 2 cores, while perturbation analyses were conducted without parallelization. As an illustration of the computational performance, the complete microscale computation for a single macrolevel material point is finished in a few seconds. Each curve

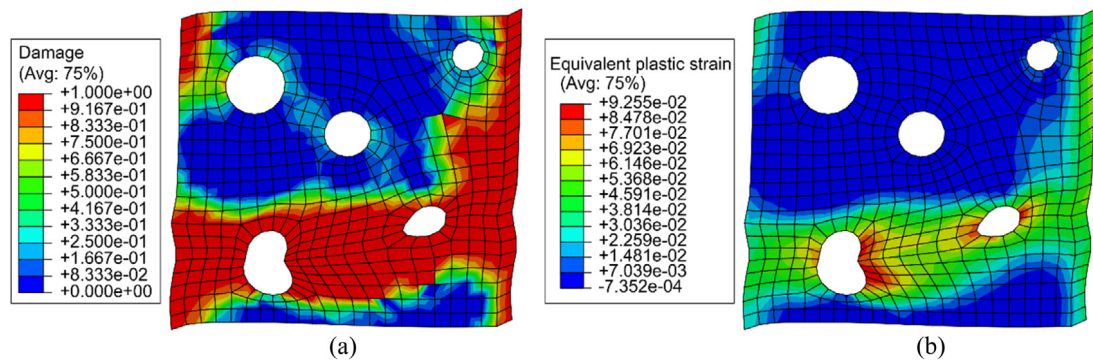


Fig. 22. Distribution of variables over DMVE at point B in Fig. 20 for loading stage II: (a) damage, (b) equivalent plastic strain.

for diagrams presented in Sections 4.2.2 and 4.2.3, where single element has been employed at the macroscale was obtained in less than two hours. However, the computation of the shear band problem consisting of 72 elements in the post-peak regime required very fine loading increments and took about five days. It should be stressed that the optimization of the computer codes can significantly reduce the computational time.

For the problems with complex geometry and loading conditions which are associated with a dense mesh discretization, the employment of the algorithms proposed requires strong hardware support. However, from engineering standpoint, the application of the multiscale algorithm in all macrolevel points of complex problems is unnecessary, bearing in mind adaptivity methods and submodelling techniques, which are available in FE software. Therefore, this algorithm can be used only in a set of elements of the macromodel, where the damage initiation and propagation are expected to occur, making the proposed procedure less time consuming and applicable in demanding problems.

6. Conclusions

The paper deals with a novel multiscale approach for modelling of damage in ductile heterogeneous materials employing the first-order computational homogenization scheme. To every macrolevel material point two microstructural models are appointed. The first one is concerned with the homogenization of the elastoplastic heterogeneous response of the microstructure including J_2 elastoplastic behaviour without softening. The second microstructural model accounts for the gradient-enhanced elastoplasticity, where the microstructural damage is governed by the nonlocal implicit formulation. The proposed microstructural softening model is implemented into the mixed quadrilateral plane strain finite element formulation, while the macrostructural discretization is performed by means of standard displacement based finite element formulation. The microstructural damage response is averaged over the area where the intensive localization is occurring. Therein a novel criterion, based on the damage increase and local equivalent strain have been proposed for the determination of loading zone. Furthermore, the classical damage law introduced at the macrolevel is assumed in a generalized form, which can also be considered as an advantage in the computation of demanding problems. Due to the generality of the macroscale damage law, the particular terms of damage partial derivatives, appearing in the constitutive macrolevel operator, are approximated by the additional DMVE computations. In other words, this makes the derived multiscale procedure applicable not only for e.g. mode I or mode II which is the usual case in available literatures but for various macrolevel loading combinations.

The macrolevel constitutive behaviour is governed by the microstructural deformation response which is computed using so-called two-step approach where the homogenization of the undamaged bulk material and the softening part has been performed separately which contributes accuracy of the state variable transition between micro and macroscale. All algorithms derived are implemented into the finite element software ABAQUS, which significantly contribute to numerical efficiency.

The new two-step scale transition procedure has been verified by means of the two well-known benchmark examples. It has been demonstrated that newly developed procedure captures evolution of the microlevel softening bands correctly, leading to the macrolevel localization paths, up to the complete failure. The presented multiscale

scheme incorporating damage is general and independent of the microscale constitutive laws. However, it must be stressed that the proposed procedure is generally applicable for proportional loading cases, where no change of loading directions occurs. When the principal directions of the stresses are not constant, anisotropic damage laws should be considered at both scales of the multiscale algorithm. This however yields more complicated numerical formulation and it is out of the scope of this paper.

It should be noted that some stability problems occur during the computation at the macrolevel which requires attentive incrementations. Therefore, in further research some additional regularization techniques should be considered in order to improve numerical efficiency and to speed up the convergence. Furthermore, the future research will be directed to the modelling of damage responses employing finite strain elastoplasticity enhanced with the gradient formulation.

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References

- [1] D. Perić, E.A. de Souza Neto, R.A. Feijóo, M. Partovi, A.J.C. Molina, On micro-to- macro transitions for multi-scale analysis of non-linear heterogeneous materials: Unified variational basis and finite element implementation, *Internat. J. Numer. Methods Engrg.* 87 (2011) 149–170, <http://dx.doi.org/10.1002/nme.3014>.
- [2] W.A.M. Brekelmans, M.G.D. Geers, V.G. Kouznetsova, Computational homogenisation for non-linear heterogeneous solids, in: *Multiscale Model. Solid Mech.*, 2009, pp. 1–42, http://dx.doi.org/10.1142/9781848163089_0001.
- [3] V.G. Kouznetsova, M.G.D. Geers, W.A.M. Brekelmans, Multi-scale second-order computational homogenization of multi-phase materials: A nested finite element solution strategy, *Comput. Methods Appl. Mech. Engrg.* 193 (2004) 5525–5550, <http://www.sciencedirect.com/science/article/pii/S0045782504002853>.
- [4] V.D. Nguyen, G. Becker, L. Noels, Multiscale computational homogenization methods with a gradient enhanced scheme based on the discontinuous Galerkin formulation, *Comput. Methods Appl. Mech. Engrg.* 260 (2013) 63–77, <http://dx.doi.org/10.1016/j.cma.2013.03.024>.
- [5] T. Lesičar, Z. Tonković, J. Sorić, Two-scale computational approach using strain gradient theory at microlevel, *Int. J. Mech. Sci.* 126 (2017) <http://dx.doi.org/10.1016/j.ijmecsci.2017.02.017>.
- [6] T. Wu, İ. Temizer, P. Wriggers, Computational thermal homogenization of concrete, *Cem. Concr. Compos.* 35 (2013) 59–70, <http://www.sciencedirect.com/science/article/pii/S0958946512001904>.
- [7] C. Miehe, Computational micro-to-macro transitions for discretized micro-structures of heterogeneous materials at finite strains based on the minimization of averaged incremental energy, *Comput. Methods Appl. Mech. Engrg.* 192 (2003) 559–591, <http://www.sciencedirect.com/science/article/pii/S0045782502005649>.
- [8] I. Gitman, H. Askes, E. Aifantis, The representative volume size in static and dynamic micro-macro transitions, *Int. J. Fract.* 135 (2005) L3–L9, <http://dx.doi.org/10.1007/s10704-005-4389-6>.
- [9] I.M. Gitman, M.B. Gitman, H. Askes, Quantification of stochastically stable representative volumes for random heterogeneous materials, *Arch. Appl. Mech.* 75 (2006) 79–92, <http://dx.doi.org/10.1007/s00419-005-0411-8>.
- [10] S. Grahame, N. Yang, Representative volumes of materials based on microstructural statistics, *Scr. Mater.* 48 (2003) 269–274, [http://dx.doi.org/10.1016/S1359-6462\(02\)00362-7](http://dx.doi.org/10.1016/S1359-6462(02)00362-7).
- [11] I.M. Gitman, H. Askes, L.J. Sluys, Representative volume: Existence and size determination, *Eng. Fract. Mech.* 74 (2007) 2518–2534, <http://dx.doi.org/10.1016/j.engfracmech.2006.12.021>.
- [12] I.M. Gitman, H. Askes, L.J. Sluys, Coupled-volume multi-scale modelling of quasi-brittle material, *Eur. J. Mech. A Solids* 27 (2008) 302–327, <http://dx.doi.org/10.1016/j.euromechsol.2007.10.004>.
- [13] T. Belytschko, S. Loehnert, J.-H. Song, Multiscale aggregating discontinuities: A method for circumventing loss of material stability, *Internat. J. Numer. Methods Engrg.* 73 (2008) 869–894, <http://dx.doi.org/10.1002/nme.2156>.
- [14] T. Hettich, A. Hund, E. Ramm, Modeling of failure in composites by X-FEM and level sets within a multiscale framework, *Comput. Methods Appl. Mech. Engrg.* 197 (2008) 414–424, <http://dx.doi.org/10.1016/j.cma.2007.07.017>.
- [15] K. Matouš, M.G. Kulkarni, P.H. Geubelle, Multiscale cohesive failure modeling of heterogeneous adhesives, *J. Mech. Phys. Solids* 56 (2008) 1511–1533, <http://dx.doi.org/10.1016/j.jmps.2007.08.005>.
- [16] T.J. Massart, R.H.J. Peerlings, M.G.D. Geers, An enhanced multi-scale approach for masonry wall computations with localization of damage, *Internat. J. Numer. Methods Engrg.* 69 (2007) 1022–1059, <http://dx.doi.org/10.1002/nme.1799>.
- [17] E. Bosco, V.G. Kouznetsova, E.W.C. Coenen, M.G.D. Geers, A. Salvadori, A multiscale framework for localizing microstructures towards the onset of macroscopic discontinuity, *Comput. Mech.* 54 (2014) 299–319, <http://dx.doi.org/10.1007/s00466-014-0986-4>.
- [18] C.V. Verhoosel, J.J.C. Remmers, M.A. Gutiérrez, R. de Borst, Computational homogenization for adhesive and cohesive failure in quasi-brittle solids, *Internat. J. Numer. Methods Engrg.* 83 (2010) 1155–1179, <http://dx.doi.org/10.1002/nme.2854>.
- [19] L.J. Sluys, V.P. Nguyen, M. Stroeve, Multiscale continuous and discontinuous modeling of heterogeneous materials: A review on recent developments, *J. Multiscale Model* 03 (2011) 229–270, <http://dx.doi.org/10.1142/S1756973711000509>.

- [20] V. Phu Nguyen, O. Lloberas-Valls, M. Stroeve, L. Johannes Sluys, On the existence of representative volumes for softening quasi-brittle materials – A failure zone averaging scheme, *Comput. Methods Appl. Mech. Engrg.* 199 (2010) 3028–3038, <http://dx.doi.org/10.1016/j.cma.2010.06.018>.
- [21] P.J. Sánchez, P.J. Blanco, A.E. Huespe, R.A. Feijóo, Failure-oriented multi-scale variational formulation: Micro-structures with nucleation and evolution of softening bands, *Comput. Methods Appl. Mech. Engrg.* 257 (2013) 221–247, <http://dx.doi.org/10.1016/J.CMA.2012.11.016>.
- [22] S. Toro, P.J. Sánchez, P.J. Blanco, E.A. de Souza Neto, A.E. Huespe, R.A. Feijóo, Multiscale formulation for material failure accounting for cohesive cracks at the macro and micro scales, *Int. J. Plast.* 76 (2016) 75–110, <http://dx.doi.org/10.1016/J.IJPLAS.2015.07.001>.
- [23] A. Ibrahimbegović, D. Marković, Strong coupling methods in multi-phase and multi-scale modeling of inelastic behavior of heterogeneous structures, *Comput. Methods Appl. Mech. Engrg.* 192 (2003) 3089–3107, [http://dx.doi.org/10.1016/S0045-7825\(03\)00342-6](http://dx.doi.org/10.1016/S0045-7825(03)00342-6).
- [24] M. Ambati, T. Gerasimov, L. De Lorenzis, Phase-field modeling of ductile fracture, *Comput. Mech.* 55 (2015) 1017–1040, <http://dx.doi.org/10.1007/s00466-015-1151-4>.
- [25] I. Benedetti, M.H. Aliabadi, Multiscale modeling of polycrystalline materials: A boundary element approach to material degradation and fracture, *Comput. Methods Appl. Mech. Engrg.* 289 (2015) 429–453, <http://dx.doi.org/10.1016/j.cma.2015.02.018>.
- [26] M. Xiang, J. Chen, Numerical simulation of ductile fracture based on mean field homogenization method: Modeling and implementation, *Eng. Fract. Mech.* 152 (2016) 147–161, <http://dx.doi.org/10.1016/J.Engfracmech.2015.07.041>.
- [27] A. Tchalla, W.L. Azoti, Y. Koutsawa, A. Makradi, S. Belouettar, H. Zahrouni, Incremental mean-fields micromechanics scheme for non-linear response of ductile damaged composite materials, *Composites B* 69 (2015) 169–180, <http://dx.doi.org/10.1016/J.Compositesb.2014.08.055>.
- [28] S. Ghosh, J. Bai, D. Paquet, Homogenization-based continuum plasticity-damage model for ductile failure of materials containing heterogeneities, *J. Mech. Phys. Solids* 57 (2009) 1017–1044, <http://dx.doi.org/10.1016/J.JMPS.2009.04.002>.
- [29] D.O. Ferdinandino, A.P. Cislino, S. Toro, P.J. Sanchez, Multi-scale analysis of the early damage mechanics of ferritized ductile iron, *Int. J. Fract.* 207 (2017) 1–26, <http://dx.doi.org/10.1007/s10704-017-0215-1>.
- [30] V. Le Corre, N. Brusselle-Dupend, M. Moreaud, Numerical modeling of the effective ductile damage of macroporous alumina, *Mech. Mater.* 114 (2017) 161–171, <http://dx.doi.org/10.1016/J.MECHMAT.2017.08.002>.
- [31] M.G.D. Geers, V.G. Kouznetsova, W.A.M. Brekelmans, Multi-scale computational homogenization: Trends and challenges, *J. Comput. Appl. Math.* 234 (2010) 2175–2182, <http://dx.doi.org/10.1016/j.cam.2009.08.077>.
- [32] K. Matouš, M.G.D. Geers, V.G. Kouznetsova, A. Gillman, A review of predictive nonlinear theories for multiscale modeling of heterogeneous materials, *J. Comput. Phys.* 330 (2017) 192–220, <http://dx.doi.org/10.1016/j.jcp.2016.10.070>.
- [33] S. Hafner, T. Luther, J. Unger, S. Eckardt, C. Konke, Multiscale simulation methods in damage prediction of brittle and ductile materials, *Int. J. Multiscale Comput. Eng.* 8 (2010) 17–36, <http://dx.doi.org/10.1615/IntJMultCompEng.v8.i1.30>.
- [34] G. Pijaudier-Cabot, Z.P. Bazant, Nonlocal damage theory, *J. Eng. Mech.* 113 (1987) 1512–1533.
- [35] E.C. Aifantis, On the role of gradients in the localization of deformation and fracture, *Internat. J. Engrg. Sci.* 30 (1992) 1279–1299, [http://dx.doi.org/10.1016/0020-7225\(92\)90141-3](http://dx.doi.org/10.1016/0020-7225(92)90141-3).
- [36] E.C. Aifantis, The physics of plastic deformation, *Int. J. Plast.* 3 (1987) 211–247, [http://dx.doi.org/10.1016/0749-6419\(87\)90021-0](http://dx.doi.org/10.1016/0749-6419(87)90021-0).
- [37] R.H.J. Peerlings, R. de Borst, W.A.M. Brekelmans, M.G.D. Geers, Gradient-enhanced damage modelling of concrete fracture, *Mech. Cohes. Frict. Mater.* 3 (1998) 323–342, [http://dx.doi.org/10.1002/\(SICI\)1099-1484\(1998100\)3:4<323::AID-CFM51>3.0.CO;2-Z](http://dx.doi.org/10.1002/(SICI)1099-1484(1998100)3:4<323::AID-CFM51>3.0.CO;2-Z).
- [38] R.H.J. Peerlings, R. De Borst, W.A.M. Brekelmans, J.H.P. De Vree, Gradient enhanced damage for quasi-brittle materials, *Internat. J. Numer. Methods Engrg.* 39 (1996) 3391–3403, [http://dx.doi.org/10.1002/\(SICI\)1097-0207\(19961015\)39:19<3391::AID-NME7>3.0.CO;2-D](http://dx.doi.org/10.1002/(SICI)1097-0207(19961015)39:19<3391::AID-NME7>3.0.CO;2-D).
- [39] R.A.B. Engelen, M.G.D. Geers, F.P.T. Baaijens, Nonlocal implicit gradient-enhanced elasto-plasticity for the modelling of softening behaviour, *Int. J. Plast.* 19 (2003) 403–433, [http://dx.doi.org/10.1016/S0749-6419\(01\)00042-0](http://dx.doi.org/10.1016/S0749-6419(01)00042-0).
- [40] R.A.B. Engelen, M.G.D. Geers, F.P.T. Baaijens, Gradient-enhanced ductile damage for the modeling of plastic deformation in softening materials, *Eur. Congr. Comput. Methods Appl. Sci. Eng.* (2000).
- [41] M.G.D. Geers, R.L.J.M. Ubachs, R.A.B. Engelen, Strongly non-local gradient-enhanced finite strain elastoplasticity, *Internat. J. Numer. Methods Engrg.* 56 (2003) 2039–2068, <http://dx.doi.org/10.1002/nme.654>.
- [42] R. De Borst, H.-B. Mühlhaus, Gradient-dependent plasticity: Formulation and algorithmic aspects, *Internat. J. Numer. Methods Engrg.* 35 (1992) 521–539, <http://dx.doi.org/10.1002/nme.1620350307>.
- [43] H.B. Mühlhaus, E.C. Aifantis, A variational principle for gradient plasticity, *Int. J. Solids Struct.* 28 (1991) 845–857, [http://dx.doi.org/10.1016/0020-7683\(91\)90004-Y](http://dx.doi.org/10.1016/0020-7683(91)90004-Y).
- [44] R. De Borst, J. Pamin, Some novel developments in finite element procedures for gradient-dependent plasticity, *Internat. J. Numer. Methods Engrg.* 39 (1996) 2477–2505, [http://dx.doi.org/10.1002/\(SICI\)1097-0207\(19960730\)39:14<2477::AID-NME962>3.0.CO;2-E](http://dx.doi.org/10.1002/(SICI)1097-0207(19960730)39:14<2477::AID-NME962>3.0.CO;2-E).
- [45] L.H. Poh, G. Sun, Localizing gradient damage model with decreasing interactions, *Internat. J. Numer. Methods Engrg.* 110 (2017) 503–522, <http://dx.doi.org/10.1002/nme.5364>.
- [46] F. Putar, J. Sorić, T. Lesičar, Z. Tonković, Damage modeling employing strain gradient continuum theory, *Int. J. Solids Struct.* 120 (2017) <http://dx.doi.org/10.1016/j.ijsolstr.2017.04.039>.
- [47] ABAQUS, ABAQUS/Standard, ABAQUS, 2014.
- [48] C. Miehe, Numerical computation of algorithmic (consistent) tangent moduli in large-strain computational inelasticity, *Comput. Methods Appl. Mech. Engrg.* 134 (1996) 223–240, [http://dx.doi.org/10.1016/0045-7825\(96\)01019-5](http://dx.doi.org/10.1016/0045-7825(96)01019-5).
- [49] A. Tchalla, S. Belouettar, A. Makradi, H. Zahrouni, An ABAQUS toolbox for multiscale finite element computation, *Composites B* 52 (2013) 323–333, <http://dx.doi.org/10.1016/j.compositesb.2013.04.028>.

- [50] Z. Yuan, J. Fish, Toward realization of computational homogenization in practice, *Internat. J. Numer. Methods Engrg.* 73 (2008) 361–380, <http://dx.doi.org/10.1002/nme.2074>.
- [51] R.J.M. Smit, W.A.M. Brekelmans, H.E.H. Meijer, Prediction of the mechanical behavior of nonlinear heterogeneous systems by multi-level finite element modeling, *Comput. Methods Appl. Mech. Engrg.* 155 (1998) 181–192, [http://dx.doi.org/10.1016/S0045-7825\(97\)00139-4](http://dx.doi.org/10.1016/S0045-7825(97)00139-4).
- [52] F. Feyel, J.-L. Chaboche, FE2 multiscale approach for modelling the elastoviscoplastic behaviour of long fibre SiC/Ti composite materials, *Comput. Methods Appl. Mech. Engrg.* 183 (2000) 309–330, [http://dx.doi.org/10.1016/S0045-7825\(99\)00224-8](http://dx.doi.org/10.1016/S0045-7825(99)00224-8).
- [53] D.J. Luscher, D.L. McDowell, C.A. Bronkhorst, A second gradient theoretical framework for hierarchical multiscale modeling of materials, *Int. J. Plast.* 26 (2010) 1248–1275, <http://dx.doi.org/10.1016/j.ijplas.2010.05.006>.